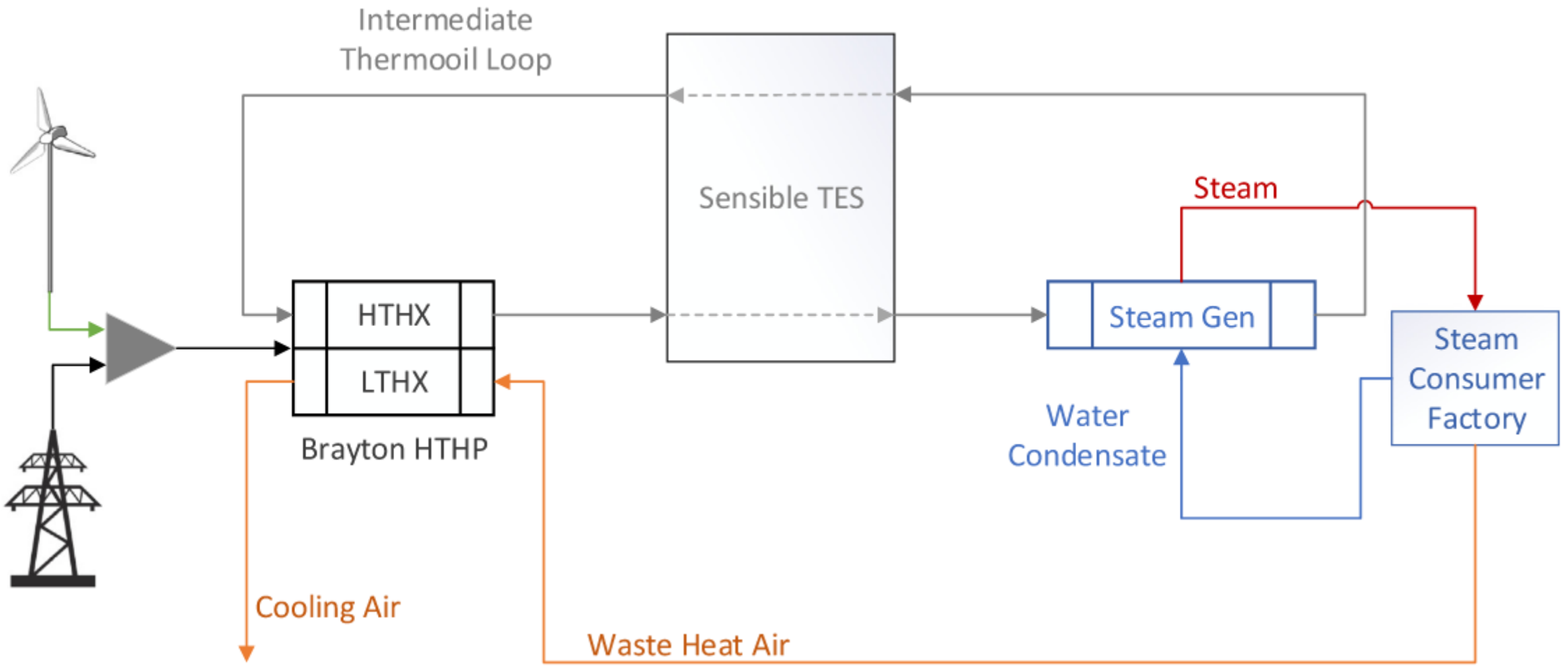
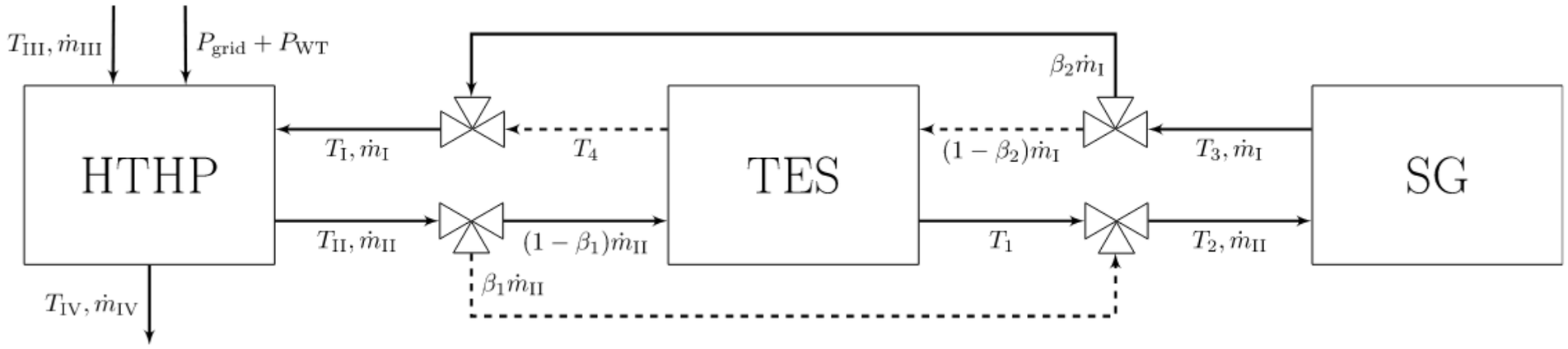


Illustration of the investigated industrial energy conversion system for electrified steam generation. The system consists of a HTHP, a TES and a SG, where the HTHP is powered by electricity from a wind turbine or the power grid. The HTHP uses waste heat air stream as a heat source and enables charging and discharging of the TES via an intermediate thermal oil stream. Furthermore, constant heat demand for the steam consumer factory must be satisfied.



Detailed flow chart of studied system including HTHP, TES and SG. HTHP and SG are represented by polynomial surrogate models, while TES is modeled by an effectiveness model. An adjustable fluid bypass β1 ∈ [0, 1] is used to control the heat input to the TES and SG, conversely the bypass β2 ∈ [0, 1] regulates the heat input to the HTHP depending on the thermal state Ts of the TES and SG outlet stream. The solid and dashed lines indicate the charging and discharging mode in a simplified way. In addition, simultaneous charging and discharging is not allowed. For example, charge mode is β1 ∈ [0, 1), β2 = 1, discharge mode implies β2 ∈ [0, 1), β1 = 1, and idle mode represents β1 = β2 = 1

Minimize total operational cost :

- **Cost depends on:**
- Electricity purchased from the grid P_t^S
- Time-varying electricity prices S_t

$$\min_x J = \sum_{t=1}^{n} P_t^S S_t \Delta t$$

Constraint-1
Power Balance Constraint

- Ensures:
  - Energy supply = energy demand
  - Total power input (grid + wind) must equal to:
    - Power required by the heat pump system
- Includes:
  - Grid power
  - Wind turbine power
  - Heat pump consumption

$$P_t^S + P_t^W = 3\,F_{HTHP}\left(T_I^t, \dot{m}_I^t, T_{III}^t, R^t\right)$$

Constraint-2,3

- These are **nonlinear surrogate models**
- They compute temperatures based on:
  - Flow rate
  - Temperatures
  - Compressor speed

$$T_{II}^k = F_{HTHX}\left(T_I^k, \dot{m}_I^k, T_{III}^k, R^k\right)$$

$$T_{IV}^k = F_{LTHX}\left(T_I^k, \dot{m}_I^k, T_{III}^k, R^k\right)$$

Constraint-4,5

- Control variables β1,β2∈[0,1]
- Define **how heat is routed** in the system

$$T_2^k = T_{II}^k \beta_1^k + T_1^k \left(1 - \beta_1^k\right)$$

➢ **Control whether energy:**

- Goes to storage
- Bypasses storage
- Returns to the heat pump

$$T_I^k = T_3^k \beta_2^k + T_4^k \left(1 - \beta_2^k\right)$$

Constraint-6,7

- Relate **temperature and mass flow rate**
- Ensure required **steam temperature conditions** are satisfied.

$$T_2^k = 201.92 + \frac{1819.32}{3\dot{m}_{II}^k}$$

$$T_3^k = 196.3 - \frac{188.4}{3\dot{m}_I^k}$$

Constraint-8,9

➢ **Thermal Energy Storage (TES)**

- Charging
- Discharging

➢ **Define heat flow into and out of storage :**

- TES stores excess heat and releases it when needed.

$$\dot{Q}_{s,ch}^{k} = 3\,\dot{m}_{II}^{k}\,c_{p,f}\left(T_{II}^{k} - T_{1}^{k}\right)\left(1-\beta_{1}^{k}\right)$$

$$\dot{Q}_{s,dch}^{k} = 3\,\dot{m}_{I}^{k}\,c_{p,f}\left(T_{4}^{k} - T_{3}^{k}\right)\left(1-\beta_{2}^{k}\right)$$

Constraint-10

➢ **No Simultaneous Charge & Discharge**

- Prevents charging and discharging at the same time

$$\dot{Q}_{s,ch}^{k}\,\dot{Q}_{s,dch}^{k} \leq \gamma$$

Constraint-11,12,13

➢ **Storage Temperature Dynamics**

- Storage temperature evolves over time (dynamic behaviour)

$$T_{1}^{k} = T_{II}^{k} - \epsilon_{ch}\left(T_{II}^{k} - T_{s}^{k-1}\right)$$

$$T_{4}^{k} = T_{3}^{k} - \epsilon_{dch}\left(T_{3}^{k} - T_{s}^{k-1}\right)$$

$$T_{s}^{k} = T_{s}^{k-1} + \frac{\dot{Q}_{s,ch}^{k} - \dot{Q}_{s,dch}^{k}}{m_{s}\,c_{p,s}}\,\Delta t$$

Constraint-14

➢ **Boundary Conditions**

- Initial and final storage temperature must match
- Ensures **cyclic daily operation**

$$T_s^0 = \widetilde{T}_s^0, \quad T_s^n = T_s^0$$

Constraint-15

➢ **Operational Limits**

$$T_I^k \in [177, 250], \quad \dot{m}_I^k \in [5, 16]$$

➢ **Limits on** :

- Temperature
- Flow rate
- Compressor speed

$$T_{III}^k \in [60, 100], \quad R^k \in [0.8, 1.53]$$

Constraint-16

➢ **Time Discretization**

- System is optimized over discrete time steps

$$t_k = \Delta t\, k, \quad k = 1, \cdots, n$$

RL formulation: observations, actions, and constraint handling

Cookie-factory SteamEnv used for the SAC evaluation

Observation o_t

- normalized TES temperature T_s^t
- current price S_t and wind availability P_t^w
- previous operating-mode hint
- 24-step price and wind forecast

Policy output and action map

$u_t = [u_\beta, u_R, u_m] \in [0,1] \times [0.8, 1.53] \times [5, 16]$

$R = u_R, \dot{m}_I = u_m$

$T_I = \beta_2 T_3 + (1 - \beta_2) T_4$

Three policy degrees of freedom determine five physical variables.

Hard feasibility handling

✓ Bypass is mapped to one valid operating mode.

$\beta_1 + \beta_2 = 1$; no simultaneous charge/discharge.

✓ Actions are clipped to the surrogate domain.

R and $\dot{m}_i$ remain within their admissible limits.

✓ Process equations determine the next state.

Reward penalties (normalized)

$r_t = -C_t - 20\, v_s - 10\, v_i - 500\, v^q - 100\, v^f - 10\, v^p$

terminal: $-1000\, v_{term}$

v_s: TES [183,324]°C;

v_i: inlet T_I [177,250]°C

v^q: thermal flow [0,5000] kW;

v^p: grid power [0,5000] kW

v^f: F4 coupling residual;

v_{term}: $|T_s^N - T_s^0|$

Hard mappings protect the action domain; penalties quantify remaining operational and terminal violations.

Cookie-factory MDP and constraints

Critical physical rule: charging and discharging the TES simultaneously is not allowed. Constraint handling therefore matters as much as the nominal cost.

Element	Definition
State	TES thermal state, electricity/wind conditions, demand and time/forecast
Actions	Compressor Speed (R_k); Mass Flow (m_I_k); bypass(beta1)
Reward	Negative electricity cost plus penalties
Constraints	Demand satisfaction, thermal-state limits, grid limits and mode coupling

Cookie-factory evaluation design

- **100** sampled 24-hour windows;
- The same exogenous data window is used for RL inference and mathematical optimisation.
- The evaluation includes post-action physical constraint checks.

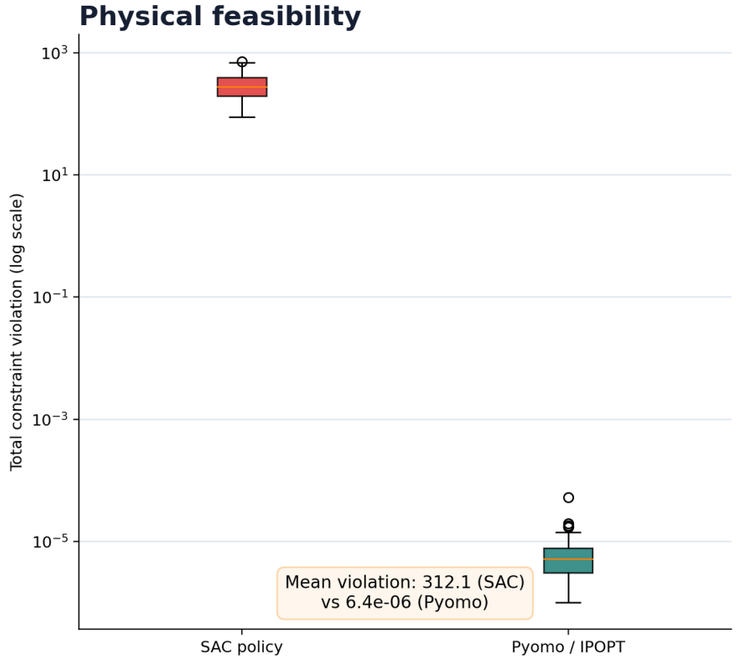


Fig:Cookie-factory case: feasibility comparison

Second Use Case: RL vs Mathematical Optimization

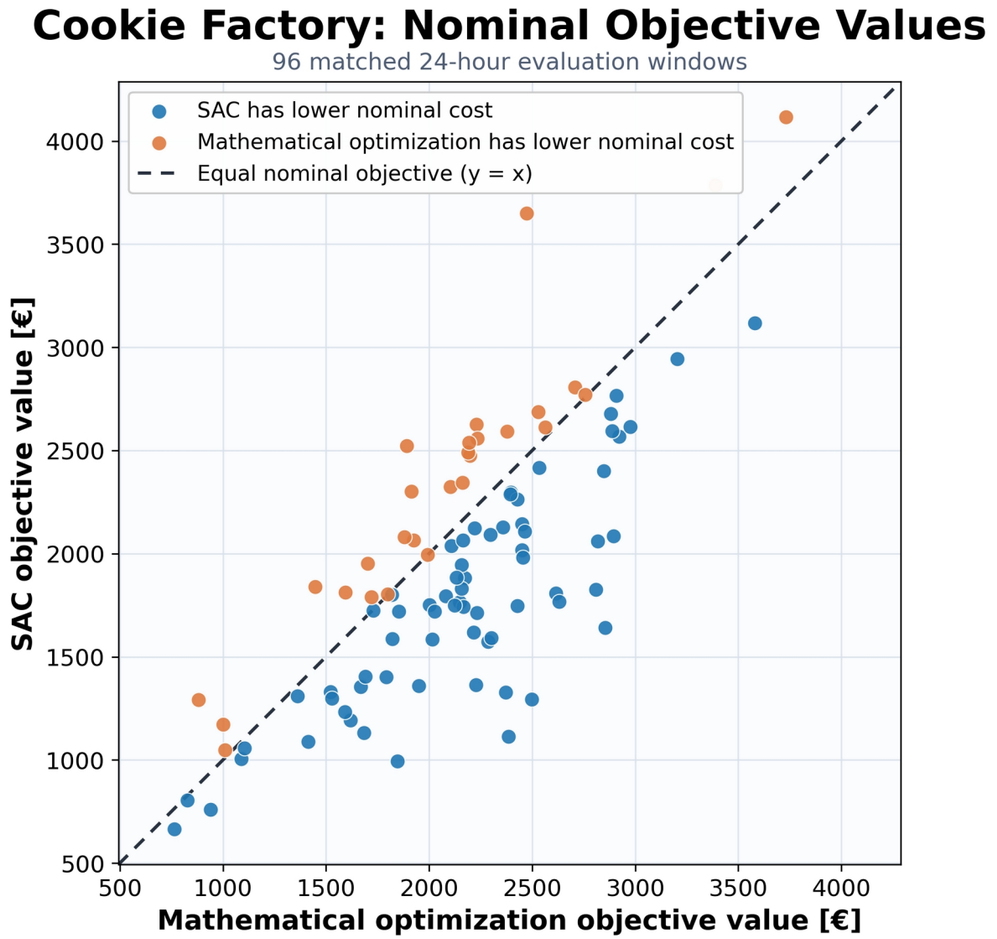
Average performance comparison for the cookie-factory energy system over 100 runs; prioritize decision time for real-time operation

Objective value (in €)

Reinforcement Learning	Mathematical Optimization
1938.44	2144.88

Objective gap: -206.44 (9.6% higher for Mathematical Optimization)

Lower nominal RL values in some windows reflect constraint violation, so they must not be interpreted as economically superior feasible schedules



Future Work: Constraint-Aware Reward Design for the Cookie Factory

1 | CURRENT LIMITATION
Nominal cost can be misleading

- SAC may achieve a low nominal electricity cost.
- However, soft penalties can permit bypass, storage, coupling, or terminal-state violations.
- A low reward cost is not a valid operational improvement unless the trajectory is feasible.

2 | PROPOSED INVESTIGATION
Strengthen the reward signal

- Add explicit, scaled violation penalties, analogous to the terminal-SOC penalty in the battery model.
- Compare fixed penalties, adaptive/dual penalties, and a feasibility-first curriculum.

3 | SUCCESS CRITERION
Feasible policy with competitive cost

- Target mean and maximum constraint violation close to zero.
- Report feasibility rate together with nominal cost and decision time.
- Benchmark only feasible SAC rollouts against mathematical optimization.

Expected outcome

A constraint-feasible SAC controller: cumulative violations → 0, while retaining as much of the learned cost and speed advantage as possible.

Thank You!

I appreciate your time and attention.